Section A For 3 or 4 set

Atomic Structure

1. Write down the postulates of Rutherford's atomic model with limitations (3/3.5)
2. Write down the postulates of Bohr's atomic model with limitations (3.75/4)
3. Derive and expression for the radius of H-atom in any orbit. (3.75/4)
4. Derive and expression for the energy of H-atom in any orbit. (3.75/4)
5. What is quantum number? Describe the four quantum number or significance of four quantum numbers. (3.5/4)
6. State and explain the term: Aufbau principle, Pauli's exclusion principle. Heisenberg uncertainty principle (3)
7. Find the four quantum number for el(15) the electron. (1.75)
8. Design the subshell for the following case.
   (i) $n = 2, l = 0$ (ii) $n = 5, l = 1$
   (iii) $n = 4, l = 3$.(1.75)

Math: rule (1) $r_n = 0.529 \times 10^{-8} cm \times n^2$ (2/2.5)

(2) $\frac{1}{\lambda} = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$ (3)

electron

↓

$m = 9.1 \times 10^{-28} g \,/\, 9.9 \times 10^{-31} kg$

$e = -1.6 \times 10^{-19} C$

↓

4.8×10^{-10} esu

(Solar system nucleus)

Rutherford's model / Postulates of Rutherford's atomic num

(1) Atom has a tiny dense center core which calle nucleus which contain the entire mass of the atom while leaving the all space almost empty.

(2) The entire positive charge of the atom is located in the nucleus while the electrons are orbiting in the outershell.

(3) The electrons are orbiting around the nucleus in a closed circular path like planets around the sun.

Limitations:

(1) The electrons were orbiting around the nucleus was unfortunate. According to the classical electromagnetic theory if two oppositely charged particle circulate the former will radiate energy and its speed will decrease and finally falling into nucleus which is not actually happend if this happend the atom will be unstable. This is the chief weakness of Rutherford's atomic model.

* Describe the postulates of Bohr's atomic model with its limitation. (3·75/4)

Ans: Bohr's atomic model was built on plancks quantum theory. According to it,

(i) Electrons are orbiting around the nucleus in specific permitted circular orbits and in each orbit electron have a definite energy and are at a fixed distance from the nucleus. The orbits are designed

$$\frac{\# Ze^2}{r^2} = \frac{mv^2}{r}$$

$$\Rightarrow \frac{Ze^2}{r} = mv^2$$

$$\Rightarrow \frac{Ze^2}{r} = m \cdot \frac{n^2h^2}{4\pi^2 m^2 r^2}$$

$$\Rightarrow Ze^2 = \frac{n^2h^2}{4\pi^2 m r}$$

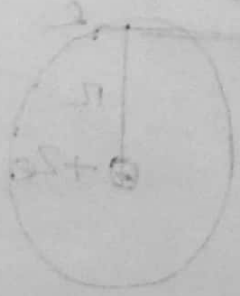
$$\Rightarrow r = \frac{n^2h^2}{4\pi^2 m Ze^2}$$

putting the value h, m, π, e we get

~~$r = \frac{n^2 h^2}{4\pi^2 m Ze^2}$~~

$$r = \frac{n^2 \times 0.529 \times 10^{-8} \text{ cm}}{Z}$$

$$\Rightarrow r_n = 0.529 \times 10^{-8} \text{ cm} \times n^2$$



Calculate the radius of He^{+} in 3rd orbit.

$$r_n = \frac{0.529 \times 10^{-8}\ cm \times n^2}{Z}$$

$$= \frac{0.529 \times 10^{-8}\ cm \times (3)^2}{2}$$

Calculate the radius of Li^{3+} ion in 2nd orbit.

$$r_n = \frac{0.529 \times 10^{-8}\ cm \times n^2}{Z}$$

$$= \frac{0.529 \times 10^{-8}\ cm \times (2)^2}{3}$$

* Derive on expression for the energy of H-atom in any orbit. OR, Calculate the energy of H-like atom in 3rd orbit.

Ans: The energy associated with revolving electron at any instant is equal to the sum of its kinetic energy(KE) and Potential Energy (P.E.). Kinetic energy of a moving electron is due to its motion in the circular orbit while the potential energy arises because the electron lies in the electric field of the positive nucleus. Again we know K.E $= \frac{1}{2}mv^2$ ——(i)

and PE $= -\frac{Ze^2}{r}$ ——(ii)

Total energy, $E = K.E + P.E$

$= \frac{1}{2}mv^2 + \left(-\frac{Ze^2}{r}\right)$

$= \frac{1}{2}\frac{Ze^2}{r} - \frac{Ze^2}{r}$ $\left[mv^2 = \frac{Ze^2}{r}\right]$

$= \frac{Ze^2 - 2Ze^2}{2r}$

$= -\frac{Ze^2}{2r}$

$= -\frac{Ze^2}{2 \times \frac{n^2h^2}{4\pi^2 mZe^2}}$ $\left[r = \frac{n^2h^2}{4\pi^2 mZe^2}\right]$

$$= -\frac{2e^2}{\frac{n^2h^2}{2\pi^2 m 2e^2}}$$

$$= \frac{-2e^2 \times 2\pi^2 m 2e^2}{n^2h^2}$$

$$= \frac{-2\pi^2 m z^2 e^4}{n^2h^2}$$

putting the value of h, π, m, e, z

$$E_n = \frac{-2.179 \times 10^{-11}}{n^2} \text{ erg/atom} \times z^2$$

Li 3rd orbit,

$$E_3 = \frac{-2.179 \times 10^{-11} \times (3)^2}{(3)^2} = -2.179 \times 10^{-11}$$

Quantum number

↓

principal / ~~principle~~ (নীতি) $(n = 1 \ldots\ldots)$

Azimuthal / Subsidary $l = (n-1)$

Magnetic $(m = \pm l)$

Spin $(s = \pm \frac{1}{2})$

↓

$l \Rightarrow 0 = s$

$\Rightarrow 1 = p$

$\Rightarrow 2 = d$

$\Rightarrow 3 = f$

সর্বোচ্চ ইলেকট্রন সংখ্যা $= 2(2l+1)$

Q. What is quantum number? Describe the four quantum number?

Or

Specification of four quantum number

Answer: Those number which are used to determine the size, shape, spin of electrons of atom are called quantum number.

There are four quantum number namely.

1. Principal quantum number
2. Azimuthal " "
3. Magnetic " "
4. Spin " "

Specification/Description the four quantum number:

1. Principal quantum number: Principal quantum number denotes the principal shell in which the electrons belongs. It is denoted by n. The value of n represent the average size of the electron cloud and n = 1, 2, 3, ... The value of n=1 means it has a one shell of lowest energy. The maximum number of electron in any orbit is $2n^2$.

* Cl (17)

Calculate the four quantum number of Cl (15th) electron.

Cl(17) → $1s^2 2s^2 2p^6 3s^2 3p^5$

$n = 3$

$l = (n-1) = 3-1 = 2$

$m = \pm l = \pm 2 = 0, \pm 1, \pm 2$

$s = \pm\frac{1}{2}, \pm\frac{1}{2}, \pm\frac{1}{2}$

Q. Given the values of l, m, s. for $n = 1, 2, 3$.

n	l	m	s	no of electrons in subshell	no of electron in orbit
1	0	0	$\pm 1/2$	2	2
2	0	0	$+\frac{1}{2}, -\frac{1}{2}$	2	8
	1	+1	$+1/2, -1/2$	2	
		0	$+1/2, -1/2$	2	
		-1	$+\frac{1}{2}, -\frac{1}{2}$	2	
3	0	0	$+\frac{1}{2}, -\frac{1}{2}$	2	18
	1	+1	$+1/2, -1/2$	2	
		0	$+\frac{1}{2}, -\frac{1}{2}$	2	
		-1	$+\frac{1}{2}, -\frac{1}{2}$	2	
	2	+2	$+\frac{1}{2}, -\frac{1}{2}$	2	
		+1	$\pm\frac{1}{2}$	2	
		0	$+\frac{1}{2}, -\frac{1}{2}$	2	
		-1	$-\frac{1}{2}, -\frac{1}{2}$	2	
		-2	$+\frac{1}{2}, -\frac{1}{2}$	2	

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$n = 1$

$l = 0$

$m = 0$

$s = +\frac{1}{2}, -\frac{1}{2}$

$n = 2$

$l = 1$

$m = -1, 0, 1$

$s = \frac{1}{2}, -\frac{1}{2} / \frac{1}{2}, -\frac{1}{2} / \frac{1}{2}, -\frac{1}{2}$

$n = 3$

$l =$

$m =$

$s =$

Q. If an electron is in 4f orbital what possible values of four quantum number of n, l, m and s it can be,

$(2l+1)$

4f

$n = 4$

$l = 3$

$m = -3, -2, -1, 0, 1, 2, 3$

$s = \pm\frac{1}{2}, \pm\frac{1}{2}, \pm\frac{1}{2}, \pm\frac{1}{2}, \pm\frac{1}{2}, \pm\frac{1}{2}, \pm\frac{1}{2}$

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Q. What is the maximum number of electron in 4d orbital. $\{2(2l+1)\}$

$s = 0$

$p = 1$

$d = 2$

$f = 3$

4d এর ক্ষেত্রে,

maximum number

$= 2(2l+1)$

$= 2(2\times2+1)$

$= 10$

Q. Which orbital is possible or not possible explain why (1p, 2s, 2p, 3f)

possible orbital = 2s, 2p

↓

Explanation, $n = 2$, $l = 0, 1$

$= 2s, 2p$

not possible orbital = 1p, 3f

↓

Explanation $n = 1$, $l = 0$

$= 1s$

↓

Explanation

$n = 3$, $l = 0, 1$

$= 3s, 3p, 3d$

Aufbauf Principle.

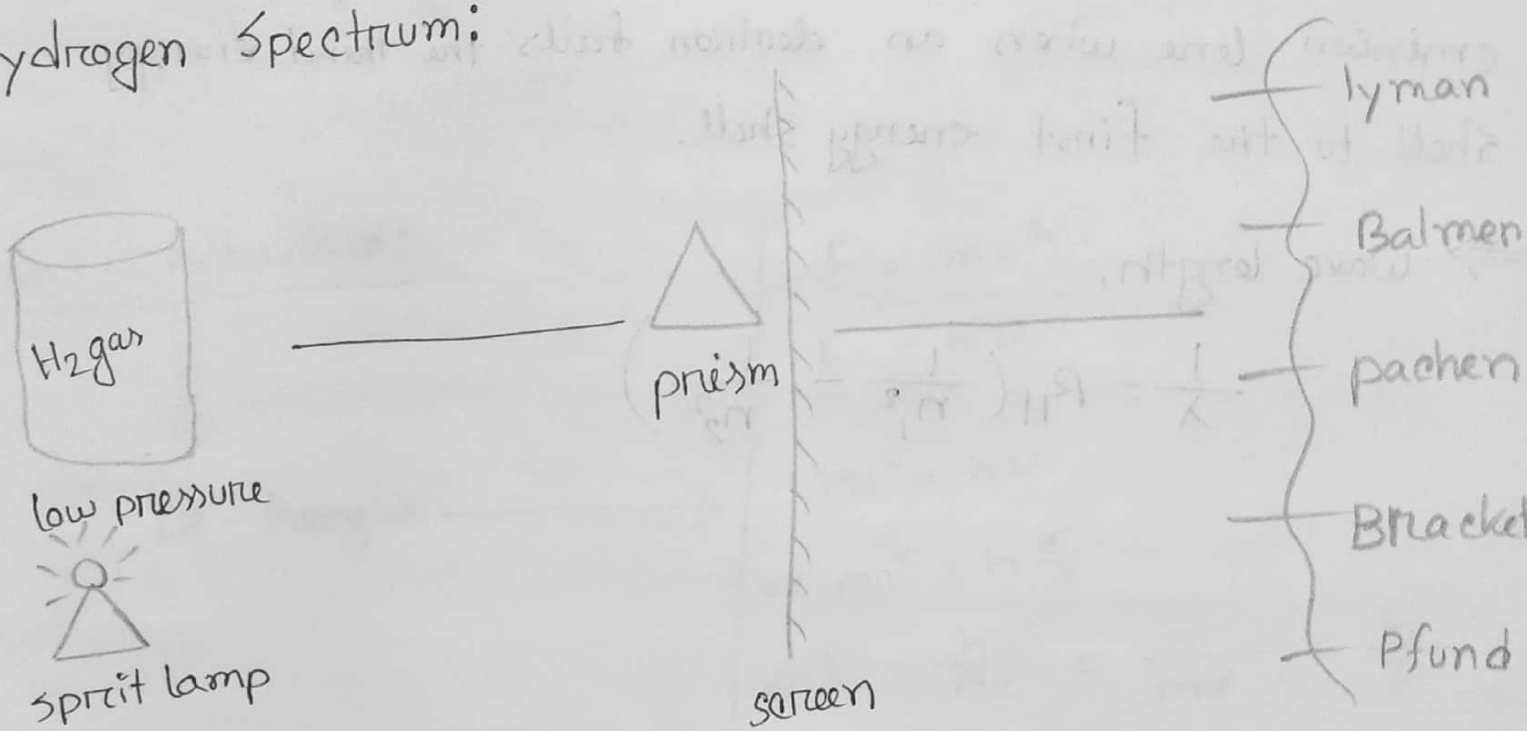
paulis exclusion principle

Hund's principle:

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02-01-23

Hydrogen Spectrum:



(i) wave length $\frac{1}{\lambda} = R_H\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

(ii) wave number $\bar{\nu} = \frac{1}{\lambda}$

(iii) frequency $\nu = \frac{c}{\lambda}$

(iv) Energy, $E = h.\nu$

$R_H = 1.097 \times 10^7\ m^{-1}$
or,
$= 1097\ cm^{-1}$

* Calculate the wavelength of lyman series.

$$\frac{1}{\lambda} = R_H\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

$$= R_H\ 1.097 \times 10^7\ m^{-1}\left(\frac{1}{(1)^2} - \frac{1}{(2)^2}\right)$$

$$=$$

m

*What is the wave length and frequency of the emission line when an electron falls the third energy shell to the first energy shell.

⇒ Wave length,

$$\frac{1}{\lambda} = R_H\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

* What are the frequency and wave length of a photon emitted during a transition from ~~n=5~~ state to the $n=2$ state in the hydrogen atom.

⇒ Wave length, $\frac{1}{\lambda} = R_H\left(\frac{1}{5^2} - \frac{1}{2^2}\right)$

$$\frac{1}{\lambda} = 1.097\left(\frac{1}{25} - \frac{1}{4}\right)$$

$$\frac{1}{\lambda} = 1.097\left(\frac{4-25}{100}\right)$$

$$\Rightarrow \frac{1}{\lambda} = \frac{1.097(-21)}{100}$$

$\frac{1}{\lambda} = 0.23037$

$\lambda = 4.34$

Frequency $\nu = \frac{1}{\lambda}$

$= 0.23037$

Dual – behaviour:

* ⇓

De - Broglie ⟶

$E = mc^2$

$E = h\nu$

$mc^2 = h\nu$

$\Rightarrow mc^2 = h\frac{c}{\lambda}$

$\Rightarrow \lambda = \frac{h}{mc} = \frac{h}{mv} = \frac{h}{p}$

2g bullet is moving at a velocity of $3\times10^2 ms^{-1}$. Calculate the wave length of the material wave associated with this bullet.

$\lambda = \frac{h}{mv}$

$= \frac{6.626\times10^{-34}}{2\times10^{-3}\times3\times10^2}$

$= \frac{6.626\times10^{-34}}{6\times10^{-1}}$

$= 1.10\times10^{-33}$

$m = 2g$

$= \frac{2}{1000} = 2\times10^{-3}kg$

$v = 3\times10^2 ms^{-1}$

$\lambda = ?$

Periodic Table

1. Write down the classification of elements on the basis of electronic configuration.

Or,

What are S and P block elements? mention the main characteristics of S and P block elements.

2. What is ionization energy?

(a) Explain why First ionization energy of AL is lower than that of Mg.

(b) Explain why first ionization energy of N is lower than that of O.

(c) First ionization potential of Na atom is less than that of Mg atom but the second ionization potential of Na atom is much more than that of Mg atom.

3. What is electron affinity? The electron affinity of chlorine is higher than that of fluorine explain why

(a) E.A of noble gas is zero or chlorine has largest electron affinity.

4. What is electro negative? Explain why F-atom is higher than that of Cl-atom. Or, Flourine is more electronegative than chlorine

5. Calculate the effective nuclear charge acting on one of the 4s electrons of V(23). Or,

Explain cations are smaller than their parent atoms / Na^+ is smaller than

Or,

Anions are larger than their Na-atom.

parents atoms / Cl^- is Larger than Cl-atom.

1. Ans: The elements are classified into 4 types on basis of electrons ~~enter into~~ configuration.

1. s-block elements: Those elements whose last electrons enter into s-orbital are called s-block elements. Ex. $Na(11) \rightarrow 1s^2 2s^2 2p^6 4s^1$

2. p-block elements: Those elements whose last electrons enter into p-orbital are called p-block elements Ex: $O(8) \rightarrow 1s^2 2s^2 2p^4$

3. d-block elements: Those elements whose last electrons enter into d-orbital are called d-block elements. Ex-

4. f-block elements: Tose elements whose last electrons ~~enter~~ into f-orbital are called f-block elements. Ex:-

I.E → The amount of energy ~~of~~ is needed to remove electrons form cation is gaseous state is called IE/IP.

$Na(g) \longrightarrow e \longrightarrow Na^+$ } $\Delta H = + = IE_1$

$Na^+ \longrightarrow e \longrightarrow Na^{++}$ } IE_2 $\quad IE_1 < IE_2$

* ioni Zation energy of Al is lower then that of Mg/ or the ionization energy of Mg is higher than that of Al.

Ans: The electronic configuration of Al and Mg

$Al_{(13)} \longrightarrow 1s^2 2s^2 2p^6 3s^2 3p_x^1$

$Mg_{(12)} \longrightarrow 1s^2 2s^2 2p^6 3s^2$

In the electronic configuration of Al the electron from $3p_x$ orbital one electron remove is easy to form a gaseous cation. But in the electronic configuration of Mg atom one electron remove from $3s^2$ orbital is difficult to form a gaseous cation. So the amount energy is high. For this reason the first ionization energy of Mg atom is higher than that of Al atom.

Another part of this question: . . . . But the second ionization energy of Al is higher than

that of Mg-atom in the electronic configuration of Al two electrons remove is difficult from the $3s^2$ orbital to form Al^{2+} ion. So the second ionization energy is high but in the electronic configuration of Mg atom two electrons remove from $3s^2$ orbital is easy So, the amount of second ionization energy is low. So, the second ionization energy of Al is higher than that of Mg atom.

* Ionization energy of O is lower than that of N or the ionization energy of N is higher that of O.

Ans:

$N_{(7)} \longrightarrow 1s^2\ 2s^2\ 2p_x^1\ 2p_y^1\ 2p_z^1$

$O_{(8)} \longrightarrow 1s^2\ 2s^2\ 2p^4$

In the electronic configuration of O the electron from $2p_x$ orbital one electron remove is easy to form a gaseous cation. But in the electronic configuration of N-atom one electron remove

from $3p_x$ orbital which is stable and difficult to remove electron from a gaseous cation. So the amount energy is high. for this reason the first ionization energy of N atom is higher than that of O atom.

Electron affinity : The amount energy released to add electron in a gaseous state to form anion is called electron affinity.

* Why electron affinity of F is lower than that of Cl.

Ans: F has lower electron affinity than Cl. Because the addition of an extra electron to F atom produce high electoron density round the resulted anion. This high electron density increases the repulsion between

the electrons already density round the ~~result~~ relatively compact 2p orbital of the 2nd shell of F-atom and the electron being added Due to the electron-electron repulsion; the F show lesser tendency to attract the extra electron from outside and hence F has lower electron affinity than Cl.

* What is electronegativity? Describe the order of electronegativity of halogens F > Cl > Br > I.
4 3.5 2.8 2.5

Ans: Electronegativity: The attraction exerted by an atom on the electron pair bonding it to another atom by a covalent bond is called electronegativity.

A ..... {:} ..... B
EN of A — E.N of B
Shared electrons

order of electronegativity (F > Cl > Br > I): In going down a group the electronegativity values decreases and going from left to right in a period the electronegativity values increases. The halogens which are lie on the extreme right of the periodic table and most electronegativity. Thus we see the most electronegative elements is F and which occurs at the top right hand corner and the least electronegative element is cesium which occurs at the

bottom left corner of the periode table. Hence the electronegativity of halogens decrease in the oder F > Cl > Br > I.

* Nuclear effective charge:

$Na(11) \rightarrow 1s^2 2s^2 2p^6 3s^1$
$\rightarrow 1s^2 2s^2 p^6 3s^1$

$= 0.35 \times 0 + 0.85 \times 8 + 1 \times 2$

$= 8.8$

$Z^* = Z - \alpha$

$= 11 - 8.8$

$= 2.2$

$\sigma = 0.35 \times$ no of electrons in (n) th shell $+ 0.85 \times$ no of electrons in (n-1) th shell + 1 x no of electron in near shell.

$Na^+ \longrightarrow 1s^2 2s^2 2p^6$
$\longrightarrow 1s^2 2s^2 p^6$

$= 0.35 \times 7 + 0.85 \times 2 + 1 \times 0$

$= 2.45 + 1.7 + 0$

$= 4.15$

$Z^* = Z - \alpha$

$= 11 - 4.15$

$= 6.85$

$Cl_{(17)} \rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$

$1s^2\ \underline{2s^2p^6}\ \underline{3s^2p^5}$

H.W ✓23

$\sigma = 0.35 \times 6 + 0.85 \times 8 + 1 \times 2$

$= 2.1 + 6.8 + 2$

$= 10.9$

$Z^* = 17 - 10.9$

$= 6.1$

$Cl < Cl^-$

$Cl^{(-)} \rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$

$1s^2\ 2s^2p^6\ 3s^2p^6$

$\sigma = 0.35 \times 7 + 0.85 \times 8 + 1 \times 2$

$= 2.45 + 6.8 + 2$

$= 11.25$

$Z^* = 17 - 11.25$

$= 5.75$

Cl^-

Anaion are larger than their parent atoms.

$V_{(23)} \longrightarrow 1s^2\ 2s^2 2p^6\ 3s^2 3p^6 4s^2 3d^3$

$1s^2,\ 2s^2p^6,\ 3s^2p^6d^3\ 4s^2,$

$\sigma = 0.35 \times 1 + 0.85 \times 18 + 1 \times 10$

$= 19.7$

$Z^* = Z - \sigma$

$= 23 - 19.7$

$= 3.3$

12-02-23

Chemical Bond

রাসায়নিক বন্ধন

1. What is chemical bond? classify chemical bond.
2. What is ionic bond? write the formation of ionic compounds.
3. What are the condition for formation of ionic bond
4. What are the properties of ionic compounds?

Explain why

(i) ionic compounds do no conduct electricity in solid state but do in liquid state.

(ii) ionic compounds are soluble in polar solvents but insoluble in non polar solvents

(iii) ionic compounds are highly brittle

5. What is covalent bond? write down the formation of covant compounds.
6. Conditions for formation of covalent compounds

7. What is Hydrogen bond? Classify H-bond.

8. What is Hybridization? Classify hybridization.

9. Shapes of the molecules: XeF_4, NH_3, SO_4^{2-}, PCl_5, ClF_3, CO_3^{2-}, BCl_3, PF_5, $BeCl_2$

10. Molecular orbital diagram and bond oder.

F_2, Cl_2, O_2, N_2, O_2^+, O_2^-

11. Explain why H_2O is liquid but H_2S in gas at NT

12. What is octet rule? Draw lewis stoucture:

PH_3, $SiCl_4$, NO, SO_4^{2-},

13. Distinguish between sigma and pi bond.

(i) ionic bond:

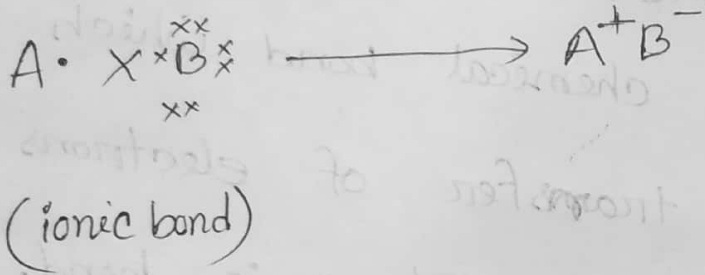
* What is chemical bond? Write down the formation of ionic bond.

Ans: The force of attraction by the atoms are liked together in a compounds is called chemical bond.

ionic bond: The chemical bond which is formed by the transfer of electrons between the metal atoms is called ionic bond. The compounds which are formed by ionic bond is called ionic compounds. Exam: NaCl, $MgCl_2$, AlC

Formation of ionic bond: Let us consider a metal atom A and a nonmetal atom B. A atom has a valence electron one and B atom has seven valence electron. A atom donate one valence electron to B atom and form A^+ and B atom

acquires one electron from A atom to form B^-. Both atom A and B acquire the stable configuration of inert gas. Thus the electrostatic force of attraction between A^+ and B^- form an ionic compound A^+B^- or AB.



(ionic bond)

27-02-23

Molecular Orbital Diagram:

$O_2 \longrightarrow$

$O + O \longrightarrow O_2$ MOT (Molecular Orbital Theory)

(8) (8)

$\longrightarrow 1s^2 2s^2 2p^4 \longrightarrow$ Energy level

$\sigma_{1s^2}\ \sigma^*_{1s^2}\ \sigma_{2s^2}\ \sigma^*_{2s^2}\ \sigma_{2p^4}\ \sigma^*_{2p^4}\ \sigma_{2p^4}\ \sigma^*_{2p^4}$

$1+1=2$

$H_2 \longrightarrow \sigma_{1s^2}\ \sigma^*_{1s^0}\ \sigma_{2s^0}\ \sigma^*_{2s^0}$

$\sigma_{1s^2}\ \sigma^*_{1s^2}$

Bond Oder

Bond Oder $=$ $\dfrac{\text{No of electron in Bonding Orbital} - \text{No of electron in antibon}}{2}$

$= \dfrac{2-0}{2} = \dfrac{2}{2} = 1$

$He_2 \longrightarrow 1s^2$

$\longrightarrow \sigma_{1s^2}\ \sigma^*_{1s^2}\ \sigma_{2s^0}\ \sigma^*_{2s^0}$

$B.O = \dfrac{2-2}{2} = \dfrac{0}{2} = 0$

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$Li_2 \longrightarrow 3+3 \rightarrow 6$

$\sigma_{1s^2} \sigma^*_{1s^2} \sigma_{2s^2} \sigma^*_{2s}$

$B.O = \frac{4-2}{2} = \frac{2}{2} = 1$

Molecular Orbital energy level ⇒

$\sigma_{1s}\ \sigma^*_{1s}\ \sigma_{2s}\ \sigma^*_{2s}\ \sigma_x \sigma_x^* \pi_y \pi_y^*\ \pi_{2p_z}\ \pi^*_{2p_z}$

$2p_y^2\ 2p_y^2$

$2p_z^2\ 2p_z$

$Cl_2 \longrightarrow 17+17(34)$

$O_2 \longrightarrow 8+8 \longrightarrow 16$

$\sigma_{1s^2}\ \sigma^*_{1s^2}\ \sigma_{2s^2}\ \sigma^*_{2s^2}\ \sigma_{2p_z^2}\ \pi^*_{2p_z^2}\ \pi_{2p_z^2}\ \pi^*_{2p_z}$

$BO = \frac{10-6}{2}$

$= \frac{4}{2} = 2$

$N_2 \longrightarrow$

$N_2 \longrightarrow 7+7 \longrightarrow 14$

$1s^2\ 2s^2\ 2p^3 \downarrow$

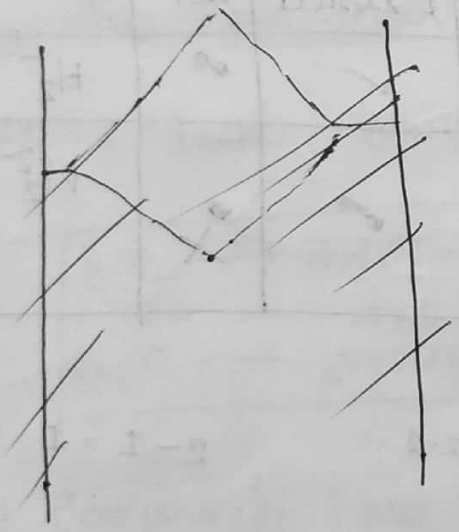
$\rightarrow 1s^2\ 2s^2\ 2p_x^1\ 2p_y^1\ 2p_z^1$

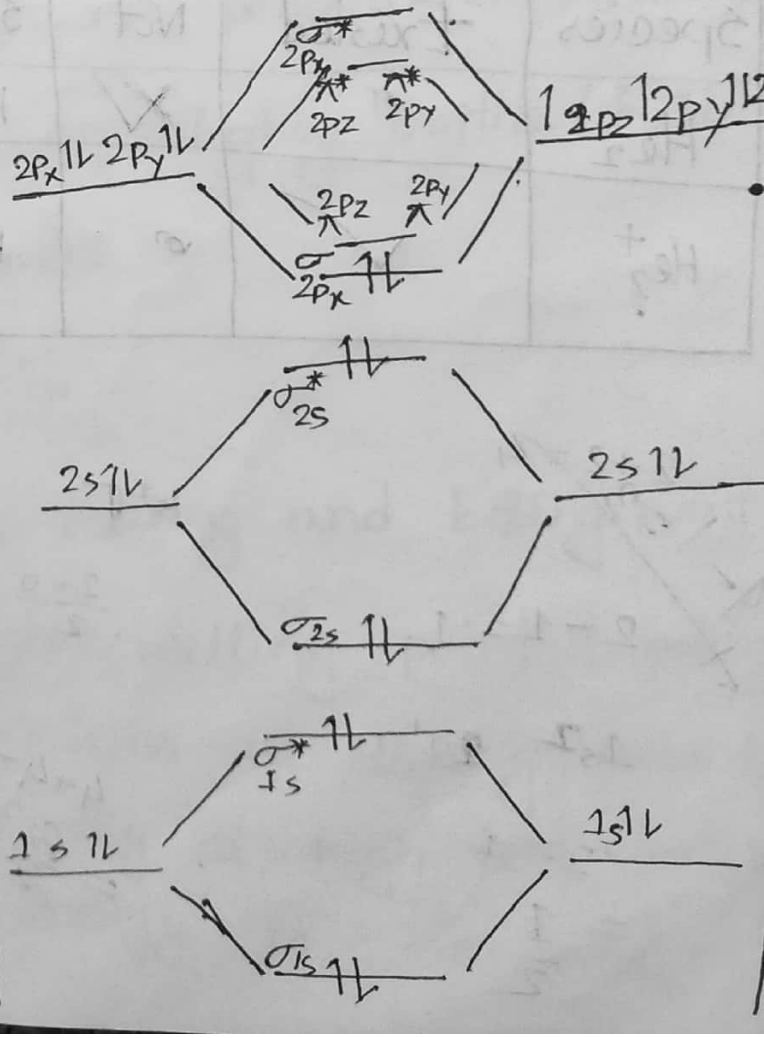
σ	σ^*	σ	σ^*	σ	σ^*	π	π^*	π	π^*
$2s^2$	$1s^2$	$2s^2$	$2s^2$	$2p_x^2$	$2p_x$	$2p_y^2$	$2p_y$	$2p_z^2$	$2p_z$

$$BO = \frac{10-4}{2} = \frac{6}{2} = 3$$

02-03-23

Molecular orbital diagram of O_2 molecule.





O = O

molecular configuration Bond Order = 2.

* Molecular orbital diagram of H_2 molecule.

* According to the molecular orbital diagram show by ticking in the boxes whether the following chemical species can be existed or not.

Species	Existed	Not	Species	Existed	Not	Species	Existed	Not
He_2		✓	He_2^{2+}	✓		H_2^-	✓	
He_2^+	✓		H_2^{2+}		✓	H_2^{2-}		✓

$2+2=4$

$4-1=3$

$2-1=1$

$1s^2\sigma$

$2+2-1=3$

$2-1=1$

$\frac{2-0}{2}=1$

$1s^2\ 2s^1$

$2-1$

$=\frac{1}{2}$

$4+4-2$

$=\frac{6}{2}=3$

$2+1=3$

$\frac{2-1}{2}$

* Write down properties of ionic compounds.

Ans: The properties of ionic compounds are given below:

(i) Physical state : Ionic compounds are three dimension at solid crystal.

(ii) Melting and boiling point: Ionic compounds have high melting and boiling point.

(iii) Electrical conductivity: Ionic compounds do not conduc electricity. But they conduct electricity in fushed or liqu or molten state.

(iv) Solubility: Ionic compounds are reading soluble in polar solvents but insoluble in non polar solvents (চাবক)

(v) Brittleness: Ionic compounds are highly brittle (ভেঙ্গুর)

CCl_4 → নন পোলার
H_2O → পোলার

* Ionic Compounds have high melting and boiling point.

Ans: Ionic compounds have high melting point & boiling points: Since the (+) and (-) ions are tightly held in their positions in the lattice, only at high temperature the ions sufficien kinetic energy to overcome

their attractive force and attains the free movement as liquid. So ionic compounds have high melting and boiling points.

* ionic compounds are soluble in polar solvents (H_2O) But insoluble in non polar (CCl_4, C_6H_6) solvents.

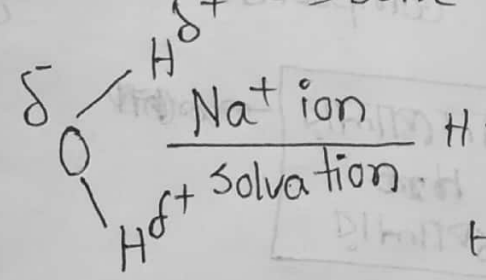
Ans: When a crystal of an ionic substance in place in water, the polar molecules datch the (+) and (-) ions form each other these ions then get surrounded by water molecules and can lead and can lead an independent existence and are thus dissolved in water. By the same reason, non polar solvents like benzene (C_6H_6) and CCl_4 will not dissolve ionic compounds

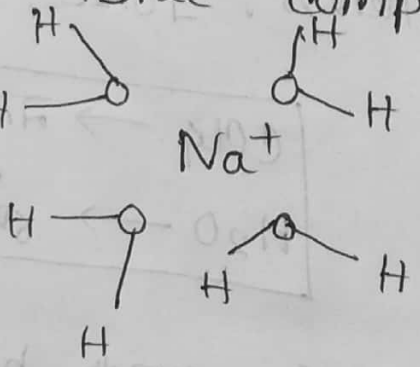
$Cl^- Na^+$

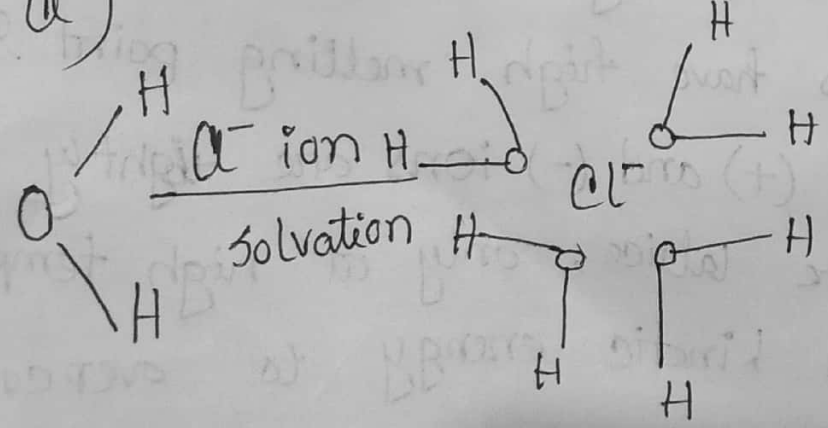
$Na^+ Cl^-$

$Cl^- Na^+$

$Na^+ Cl^-$







1. Explain why ionic compounds do not conduct electricity in solid state but do in liquid or molten state.

Ans: Ionic compounds do not conduct electricity in solid state because no current flow in the solid because ions are immobile for the strong electrostatic force between the positive ions and negative ions. But in molten state the positive and negative ions (mobile ions) flow toward the oppositely charge electrodes and carry current. So it is said that in aqua solution or liquid or molten state ionic compounds conduct electricity.

2. Explain why Ionic compounds are highly brittle

Ans: The positive and negative ions in the crystal are arranged to maximized their attractions when an external force is applied, like charges move near each other and the repulsion crack the piece crack. So, ionic compounds are highly brittle.

* What are the condition for formation of ionic compounds or What condition should be fullfilled to form an ionic bond between A & B atom.

Ans: Condition for formation of ionic bond are given below:

(i) Number of valance electron: The number valance electron of metal atom(A) should be 1,2,3 and the number of valance electrons of nonmetal atom(B) should be 7,6,5.

(ii) The ionization energy: The ionization energy of metal atom should be low. $A_{(g)} \longrightarrow e^- \longrightarrow A^+_{(g)}$

(iii) The electron affinity: The electron affinity of nonmetal atom should be high.

$$B_{(g)} + e \longrightarrow B^-_{(g)}$$

(iv) The electronegativity: The electronegativity difference between the metal and non metal atom should be high or greater than 2. (Ex: Na = 0.9, Cl = 3, so, NaCl = 3.0 - 0.9 = 2.1)

(v) Lattice Energy: The lattice energy of ionic compound should be high.

$$A^+ + B^- \longrightarrow A^+B^- + \text{Energy level} \downarrow$$

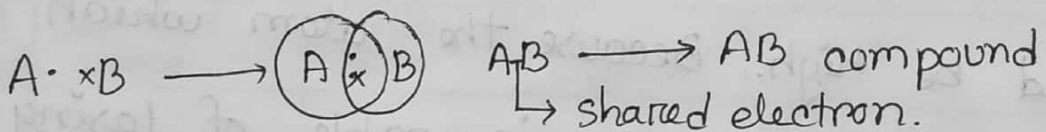
lattice Energy.

23-02-23

*What is covalent bond?

The chemical bond which is formed between the two atom by sharing one or more electron pair is called covalent bond. Ex: F_2, Cl_2, H_2O, N_2.

Formation of covalent bond: Let us consider two nonmetal atom A & B. A atom has one valance electron and B atom has one valance electron. Both atom acquire the inert gas configuration by sharing one electron. A atom and B atom contribute one electron to accure stable configuration By sharing one electron both atom form a bond which is termed as covalent bond.



* Properties of covalent bond/compounds.

i. Covalent compound are discrete giant molecules.

ii. Covalent compounds are soluble in non polar solvent and insoluble in polar solvents.

iii. They are bad conductors of heat and electricity.

iv: Covalents compounds are neither hard or brittle.

v. They have low melting and boiling point.

*Distingwish between the properties ionic and covalent compounds.

1. Condition for formation of covalent bond

Ans: Ionization energy: The ionization energy of the atom should be high. Because the atom which have high ionization energy are incapable of losing electrons to form cations. So they form covalent bond by having high ionization energy.

Electron affinity: The electron affinity of the non metal atom should be equal ~~because~~ because the

non metal atom ~~should be~~ have equal attraction electrons.

Number of valance electron: Each two atoms should hav 5, 6, 7. [except H] by sharing 3, 2, 1 electron pair.

Electronegativity: The electronegativity of the two atom should be equal.

06-03-2023

Hybridization:

Region of electron density	Geometric Shap (arrangement)	Hybridization name	Bond angle	Example
2	Linear	sp	180°	$BeCl_2$, $HgCl_2$, CO_2
3	Trigonal	sp^2	120°	BF_3, CO_3^{2-}, NO_3^-, SO_3
4	Tetrahedral	sp^3	109.5°	CH_4, PO_4^{3-}, SO_4^{2-}, ClO_4^-, $TiCl_4$, X_2O_4
5	Trigonal bipiramical	sp^3d	90° and 120°	PCl_5
6	Octahedral	sp^3d^2	90° and 90°	SF_6

$H = \frac{1}{2}(V+M-C+A)$ (যোজ্যতা ইলেকট্রন)

$= \frac{1}{2}$ (কেন্দ্রীয় পরমাণুর জারণ মান + একযোজী পরমাণুর সংখ্যা

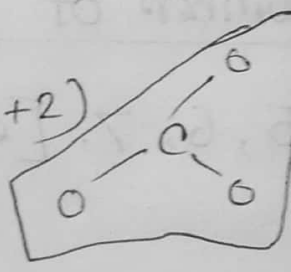
− ক্যাটায়ন সংখ্যা + অ্যানায়ন সংখ্যা)

$BeCl_2 = \frac{1}{2}(2+2-0+0)$

$= \frac{1}{2} \times 4 = 2$

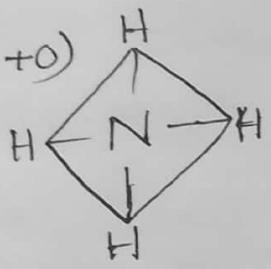
$CO_3^{2-} = \frac{1}{2}(4+0-0+2)$

$= \frac{1}{2} \times 6 = 3$



$NH_4^+ = \frac{1}{2}(5+4-1+0)$

$= 4$



$SF_6 = \frac{1}{2}(6+6-0+0)$

$= 6$



Electrochemistry

1. Describe the Faraday's law of electrolysis. Describe the significance of this laws.
2. State and explain kohlrausch's law of independent ion migration.
3. Describe the calomel referance electrode and give half-well reaction when it is coupled with a standard hydrogen electrode.
4. Debye-Hukel onsager theory of strong electrolyte.
5. Define the term: strong electrolyte, ionic mobilety, transpart number, mettalic and electrolytic conduction salt bridge, electromotive force, specific conductance, cell constant, molar conductance.
6. What is primary and seoondary refference electrode. Describe the construction and working principle of hydrogen electrode.
7. What is salt bridge? Describe function of it.
8. Distinguish between galvanic cell and electrolytic cell.
9. Explain why,
   (i) The equivalent conductance of a soletion of strong electrolyte gradually increases.

(ii) The specific conductance of weak electrolyte decreases rapidly with increase in concentration

10. What happends when a copper rod is dipped in a Zinc sulphate solution.

Math:

(i) $E^{\circ} = E - \frac{0.0591}{n} \log \frac{[\text{Oxidation}]}{[\text{Reduction}]}$

(ii) $W = ZQ$

(iii) Kohaurch's law.

(iv) Spontaneity of reaction.

Electrolyte Cell:

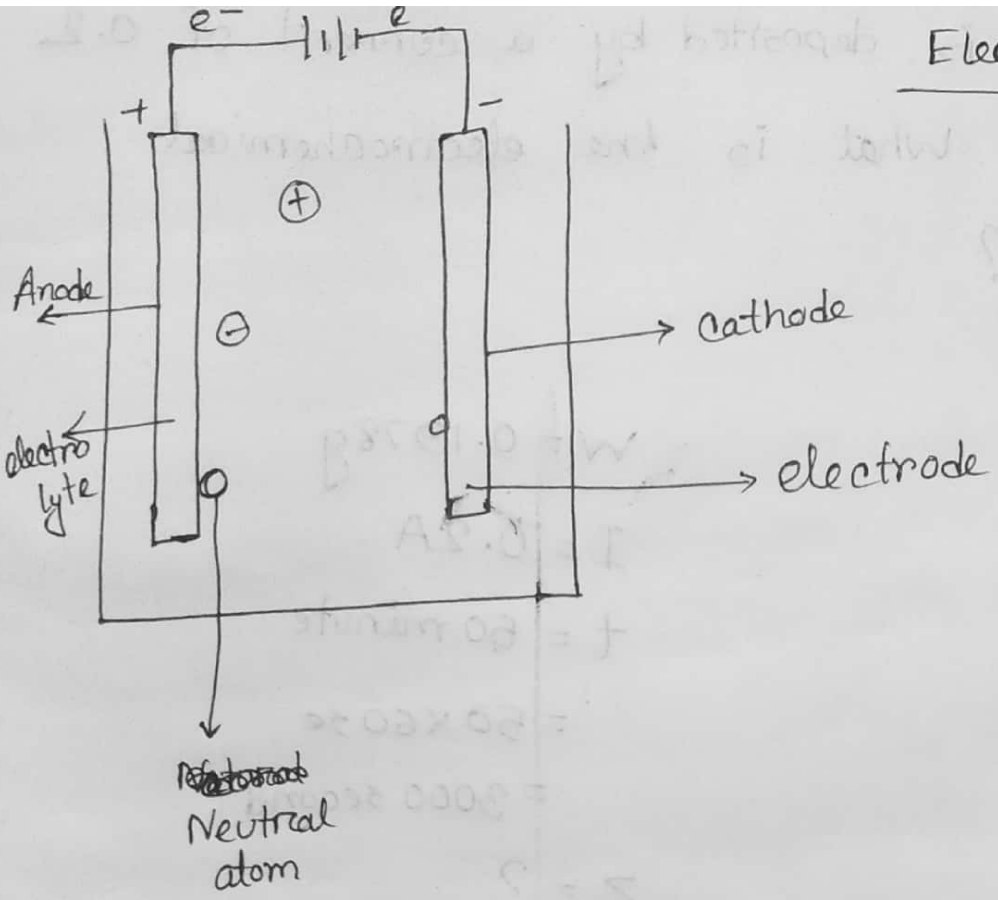


Fig: Electrolysis

Anode reaction: $Cl^- \longrightarrow Cl + e^-$ (oxidation)

Cathode reaction: $H^+ + e \longrightarrow H$ (Reduction)

Overall reaction: $2HCl \longrightarrow H_2 + Cl_2$

Faraday's law of electrolysis: ① $w \propto Q$

$w = zQ$

$= z.It$

② $W_A \propto Z_1$, $W_B \propto Z_2$, $W_C \propto Z_3$

$W_A = KZ_1$, $W_B = KZ_2$, $W_C = KZ_3$

$$\frac{W_A}{Z_1} = \frac{W_B}{Z_2} = \frac{W_C}{Z_3}$$

0.1978g of copper is deposited by a current of 0.2 ampere in 50 minute. What is the electrochemical equivalent of copper?

$W = ZIt$

$\Rightarrow Z = \frac{W}{It}$

$= \frac{0.1978}{0.2 \times 3000}$

$= 0.0003296$

$W = 0.1978g$

$I = 0.2A$

$t = 50$ minute

$= 50 \times 60$ se

$= 3000$ second.

$Z = ?$

*What current strength in ampere will be required to liberate 30g iodine from potassium iodide solution in one hour?

$\Rightarrow$

$W = 30g$

$t = 1$ hour

$= 60 \times 60 = 3600$ second

$I = ?$

Quantity of electricity $= n \times F$

$= n \times 96500$